

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**1.1. Product identification**

Product Code/Catalogue Number: 984309
SDS Number: D14456_SDS_Tartaric Acid R1 _EN
Product Name **Tartaric Acid R1**

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.

1.3. Details of the supplier of the safety data sheet

Company **Thermo Fisher Scientific Oy**
Ratastie 2,
FI-01620 Vantaa, Finland
Telephone number +358 10 329200
E-mail address system.support.fi@thermofisher.com

1.4. Emergency telephone number

CHEMTREC INTERNATIONAL +1 703-741-5970

SECTION 2: HAZARDS IDENTIFICATION**2.1. Classification of the substance or mixture****CLP Classification - Regulation (EC) No 1272/2008**

Substances/mixtures corrosive to metal	Category 1 (H290)
Skin Corrosion/irritation	Category 1 A (H314)
Serious Eye Damage/Eye Irritation	Category 1 (H318)

2.2. Label elements

Signal Word

Danger

Hazard Statements

H290 - May be corrosive to metals

H314 - Causes severe skin burns and eye damage

Precautionary Statements

P280 - Wear protective gloves/ eye protection/ face protection

P303 + P361 + P353 - IF ON SKIN (or hair): Remove/ Take off immediately all contaminated clothing. Rinse skin with water/ shower

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

2.3. Other hazards

No information available

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS**3.2. Mixtures**

Component	Weight %	CLP Classification - Regulation (EC) No 1272/2008
Sulfuric acid (CAS #: 7664-93-9)	1 - < 5 %	Skin Corr. 1A (H314) Eye Dam. 1 (H318)
Tartaric acid (d, l) (CAS #: 87-69-4)	0.01 - < 0.1 %	Eye Dam. 1 (H318)

Component	Reach Registration Number	
Sulfuric acid	01-2119458838-20-xxxx	
Tartaric acid (d, l)	01-2119537204-47-xxxx	

*Full text of Hazard Statements: see section 16***SECTION 4: FIRST AID MEASURES****4.1. Description of first aid measures****General Advice**

Consult a physician. Show this safety data sheet to the doctor in attendance.

Inhalation

Move to fresh air. Get medical attention if symptoms occur.

Skin Contact

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get medical attention if symptoms occur.

Eye Contact

Rinse thoroughly with plenty of water for at least 15 minutes and consult a physician.

Ingestion

Clean mouth with water and drink afterwards plenty of water. Do not induce vomiting. Never give anything by mouth to an unconscious person. Rinse mouth. Consult a physician.

4.2. Most important symptoms and effects, both acute and delayed

No information available.

4.3. Indication of any immediate medical attention and special treatment needed

Treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES**5.1. Extinguishing media****Suitable Extinguishing Media**Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. alcohol-resistant foam. Dry chemical. Carbon dioxide (CO₂).**Extinguishing media which must not be used for safety reasons**

No information available.

5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

Hazardous Combustion Products

None under normal use conditions.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES**6.1. Personal precautions, protective equipment and emergency procedures**

Ensure adequate ventilation. Use personal protective equipment. Evacuate personnel to safe areas.

6.2. Environmental precautions

Prevent further leakage or spillage if safe to do so. Do not flush into surface water or sanitary sewer system.

6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE**7.1. Precautions for safe handling**

Ensure adequate ventilation. Remove all sources of ignition.

7.2. Conditions for safe storage, including any incompatibilities

Keep container tightly closed in a dry and well-ventilated place. Keep at temperatures between 2° and 8 °C.

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**8.1. Control parameters****Component Exposure Limits**

Component	Finland	European Union	The United Kingdom	Germany
Sulfuric acid	TWA: 0.05 mg/m ³ 8 tunteina STEL: 0.1 mg/m ³ 15 minuutteina	TWA: 0.05 mg/m ³ (8h)	STEL: 0.15 mg/m ³ 15 min TWA: 0.05 mg/m ³ 8 hr	TWA: 0.1 mg/m ³ (8 Stunden). AGW - exposure factor 1 TWA: 0.1 mg/m ³ (8 Stunden). MAK Höhepunkt: 0.1 mg/m ³
Tartaric acid (d, l)				TWA: 2 mg/m ³ (8 Stunden). AGW - exposure factor 2 TWA: 2 mg/m ³ (8 Stunden). MAK Höhepunkt: 4 mg/m ³

Component	Sweden	Norway	Denmark	France
Sulfuric acid	Indicative STEL: 0.2 mg/m ³ 15 minuter TLV: 0.1 mg/m ³ 8 timmar. NGV	TWA: 0.1 mg/m ³ 8 timer STEL: 0.3 mg/m ³ 15 minuter. value calculated thoracic fraction	TWA: 0.05 mg/m ³ 8 timer	TWA / VME: 0.05 mg/m ³ (8 heures). STEL / VLCT: 3 mg/m ³ .

8.2. Exposure controls**Engineering Measures**

Ensure adequate ventilation, especially in confined areas.

Personal protective equipment

Eye Protection Safety glasses with side-shields (European standard - EN 166)

Hand Protection Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Disposable gloves	See manufacturers recommendations	-	EN 374	(minimum requirement)

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

Skin and body protection

Long sleeved clothing

Respiratory Protection When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

When RPE is used a face piece Fit Test should be conducted

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

Environmental exposure controls

No information available.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Appearance	No information available	
Physical State	Liquid	
Odor	Characteristic	
Odor Threshold	No data available	
pH	< 2.0	
Melting Point/Range	No data available	
Softening Point	No data available	
Boiling Point/Range	100 °C	
Flash Point	No data available	Method - No information available
Evaporation Rate	No data available	
Flammability (solid,gas)	No information available	
Explosion Limits	No data available	
Vapor Pressure	No data available	
Vapor Density	No data available	(Air = 1.0)
Specific Gravity / Density	No data available	
Bulk Density	No data available	
Water Solubility	soluble	
Solubility in other solvents	No information available	

Partition Coefficient (n-octanol/water)

Component	log Pow
Tartaric acid (d, l)	-1.7
Autoignition Temperature	No data available
Decomposition Temperature	No data available
Viscosity	No data available
Explosive Properties	No information available
Oxidizing Properties	No information available

9.2. Other information

No data available

SECTION 10: STABILITY AND REACTIVITY**10.1. Reactivity**

No data available

10.2. Chemical stability

Stable under normal conditions

10.3. Possibility of hazardous reactions

None under normal processing.

10.4. Conditions to avoid

No information available.

10.5. Incompatible materials

None known.

10.6. Hazardous decomposition products

None under normal use conditions.

SECTION 11: TOXICOLOGICAL INFORMATION**11.1. Information on toxicological effects****Product Information**

No acute toxicity information is available for this product

(a) acute toxicity;**Oral** Not Classified**Dermal** Not Classified**Inhalation** Not Classified

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Sulfuric acid	2140 mg/kg (Rat)		LC50 85 - 103 mg/m ³ (Rat) 1 h

(b) skin corrosion/irritation;

Category 1. A.

(c) serious eye damage/irritation;

Category 1.

(d) respiratory or skin sensitization;

Respiratory

Not Classified.

Skin

Not Classified.

(e) germ cell mutagenicity;

Not Classified

(f) carcinogenicity;

Not Classified

There are no known carcinogenic chemicals in this product

Component	EU	UK	Germany	IARC
Sulfuric acid				Group 1

(g) reproductive toxicity;

Not Classified.

(h) STOT-single exposure;

Based on available data, the classification criteria are not met.

(i) STOT-repeated exposure;

Not Classified.

Target Organs

No information available.

(j) aspiration hazard;

Not Classified.

Symptoms / effects, both acute and delayed

No information available

SECTION 12: ECOLOGICAL INFORMATION**12.1. Toxicity**

Component	Freshwater Fish	Water Flea	Freshwater Algae	Microtox
Sulfuric acid	LC50: > 500 mg/L, 96h static (Brachydanio rerio)	EC50: 29 mg/L/24h	-	-
Tartaric acid (d, l)	-	EC50=230 mg/L 48h	-	-

12.2. Persistence and degradability

No information available

12.3. Bioaccumulative potential

No information available

Component	log Pow	Bioconcentration factor (BCF)
Tartaric acid (d, l)	-1.7	No data available

12.4. Mobility in soil

No information available

12.5. Results of PBT and vPvB assessment

No data available for assessment.

12.6. Other adverse effects

None known

SECTION 13: DISPOSAL CONSIDERATIONS**13.1. Waste treatment methods****Waste from Residues / Unused Products**

Dispose of in accordance with local regulations.

Contaminated Packaging

Dispose of in accordance with local regulations.

SECTION 14: TRANSPORT INFORMATION

	IMDG/IMO	ADR	IATA
14.1. UN number	UN1760	UN1760	UN1760
14.2. UN proper shipping name	CORROSIVE LIQUIDS, N.O.S; SULPHURIC ACID SOLUTION; SODIUM HYDROXIDE SOLUTION	CORROSIVE LIQUID, N.O.S; SULPHURIC ACID SOLUTION; SODIUM HYDROXIDE SOLUTION	CORROSIVE LIQUIDS, N.O.S; SULPHURIC ACID SOLUTION; SODIUM HYDROXIDE SOLUTION
14.3. Transport hazard class(es)	8	8	8
14.4. Packing group	III	III	III

14.5. Environmental hazards

No hazards identified

14.6. Special precautions for user

No special precautions required

14.7. Transport in bulk according to Annex II of MARPOL73/78 and the IBC Code

Not applicable, packaged goods

SECTION 15: REGULATORY INFORMATION

This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories X = listed

Component	EINECS	ELINCS	NLP	TSCA	DSL	NDSL	PICCS	ENCS	IECSC	AICS	KECL
Sulfuric acid	231-639-5	-		X	X	-	X	X	X	X	KE-32570
Tartaric acid (d, l)	201-766-0	-		X	X	-	X	X	X	X	KE-10801

National Regulations

Component	Germany - Water Classification (VwVwS)	Germany - TA-Luft Class
Sulfuric acid	WGK1	

Tartaric acid (d, l)	WGK1	
----------------------	------	--

15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3

H290 - May be corrosive to metals

H314 - Causes severe skin burns and eye damage

H318 - Causes serious eye damage

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

PNEC - Predicted No Effect Concentration

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - Volatile Organic Compounds

Key literature references and sources for data

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Version

2

Revision Date

10-Sep-2019

Reason for revision

SDS section(s) updated: 1, 3, 14, 16.

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text